

INTRODUCTION TO ASP.NET

Web Frameworks

ASP.NET is one of the early web application frameworks that allows building dynamic of websites. ASP stands for “Active Server Pages”, which was Microsoft's initial web application framework that used server-side processing to make browser readable HTML pages.

ASP.NET extends the .NET platform with tools and libraries for creating web applications. Listed below are ASP.NET's features:

- Base framework for processing web requests in C# or F#
- Web-page templating syntax, known as Razor, for building dynamic web pages using C#
- Libraries for common web patterns, such as Model View Controller (MVC)
- Authentication system that includes libraries, a database, and template pages for handling logins, including multi-factor authentication and external authentication with Google, Twitter, and more
- Editor extensions to provide syntax highlighting, code completion, and other functionality specifically for developing web pages

ASP.NET was considered a major enhancement since it allows much cleaner code-behind. Compared to the other frameworks, it has the code that handles processing and markup and building of display.

ASP.NET allows you to use a full-featured programming language such as C#.NET or VB.NET.

There are three (3) frameworks for creating web applications: Web Forms, ASP.NET MVC, and ASP.NET Web Pages.

- **Web Forms** - enables building of dynamic websites using a drag-and-drop, event-driven model. With its hundreds of controls and components, you can build sophisticated, powerful UI-driven sites with data access.
- **ASP. NET MVC** – gives a powerful, patterns-based way to build dynamic websites that gives you full control over markup for agile development. It includes many features that enable fast, TDD-friendly development for creating sophisticated applications that use the latest web standards.
- **ASP.NET Web Pages** – provides a fast and lightweight way to combine server code with HTML to create dynamic web content. You can connect to databases, add video, link to social networking sites, and many more ways to help you create beautiful sites that conform to the latest web standards.

Bootstrap is a web development framework that helps developers in building web interfaces. The framework consists of a mixture of JavaScript and CSS. It provides developers with all the essential components required to develop a fully-functioning web user interface.

jQuery is a JavaScript library that allows HTML document traversal and manipulation, event handling, and animation.

AngularJS is a powerful JavaScript framework developed by Google. AngularJS provides easily consumable abstractions on top of JavaScript to aid the development of web applications.

HTTP and HTTPS

Hypertext Transfer Protocol (HTTP) is an application-layer protocol that acts as the foundation for communications with the Internet. It defines the interaction between the client machine and the server as following a request-response model whereby the client machine requests for a specific resource and the server responds with the information.

HTTPS stands for Hypertext Transfer Protocol Secure. It is the secure version of HTTP. It is encrypted in order to increase security. The following are the most commonly used HTTP methods.

Name	Description
GET	It retrieves the source without any other effect resulting from taking actions.
POST	It indicates that there is information included in the request that should create a new version of the resource.

PUT	It indicates that the information included in the request should change already existing items.
DELETE	The specified resource should be deleted. And upon deletion, the GET or PUT to a resource will fail.

IIS stands for Internet Information Services, an application that comes with Microsoft Windows designed to support HTTP (known as a web server). When developing ASP.NET applications, either Web Forms or MVC, the work of processing and creating the content that is returned to the client is done by IIS.

Visual Studio

Visual studio is the primary integrated development environment (IDE) used to create ASP.NET websites and web applications.

There are three (3) versions that can be downloaded: Community, Professional, and Enterprise.

- Community – free for students, open-source contributors, and individuals
- Professional – best suited to small teams
- Enterprise – scalable, end-to-end solution for teams of any size

REFERENCES:

Andrew, L. (2021). *ASP.NET Core in action*. Manning Publications Co.

Benjamin J. & Jason M. (2018). *Mastering Bootstrap 4 2nd Edition: Master the latest version of Bootstrap 4 to build highly customized responsive web apps*. Packt Publishing.

De Oliveria, J. & Bruchet, M. (2017). *Learning ASP.NET Core 2.0*. Packt Publishing.

Rippon, C. (2020). *ASP.NET Core 5 and React: Full-stack web development using .NET 5, React 17, and TypeScript 4*. Packt Publishing.